
Problem 1 (Real numbers). *Answer the following questions. Show all necessary steps and reasoning.*

1. *Prove that every integer is a real number.*
2. *Show that the sum of two irrational numbers is irrational.*
3. *Prove that the product of a nonzero rational number and an irrational number is irrational.*
4. *Determine whether $\sqrt{2}$ is a rational or irrational number. Justify your answer.*
5. *Prove that the square root of any prime number is irrational.*
6. *Given $a, b \in \mathbb{R}$, prove that $|a + b| \leq |a| + |b|$.*
7. *Define the absolute value function $|x|$ for $x \in \mathbb{R}$.*
8. *Show that for any real numbers a and b , $|ab| = |a| \cdot |b|$.*
9. *Prove that the set of rational numbers is dense in the set of real numbers.*
10. *Define a Cauchy sequence. Provide an example of a Cauchy sequence that converges to an irrational number.*
11. *Show that the square root of any positive real number is a real number.*
12. *Prove that if a and b are rational numbers such that $a < b$, then there exists an irrational number between a and b .*
13. *Define a real number as a limit of a convergent sequence. Provide an example.*
14. *Show that the set of real numbers is uncountably infinite.*
15. *Prove or disprove: The sum of a rational number and an irrational number is always irrational.*

Problem 2 (Inequality). 1. *Solve the inequality: $3x - 5 > 2x + 8$.*

2. *Determine the solution set for the inequality: $-2(4x + 1) \leq 3(2x - 5)$.*
3. *Solve the compound inequality: $-3 < 2x + 4 \leq 5$.*
4. *Find the values of x that satisfy the inequality: $\frac{2}{x} \geq 4$.*
5. *Solve the absolute value inequality: $|3x - 2| < 7$.*
6. *Determine the solution set for the inequality: $2x^2 - 5x + 3 < 0$.*

7. Solve the system of inequalities:

$$\begin{aligned}2x + y &\leq 5 \\ x - 3y &> 2\end{aligned}$$

8. Solve the following inequality

$$\frac{x+4}{x+8} > \frac{x-6}{2x-3}$$

9. Given that $a - 2b > 4$ and $3a + b \leq 5$, find the possible values for a and b .

10. Solve the quadratic inequality: $x^2 - 4x + 3 > 0$.

Problem 3 (Modulus). 1. Determine the solution set for the modulus equation: $|2x + 1| = 5$.

2. Solve the modulus inequality: $|3x - 2| \geq 4$.

3. Find the values of x that satisfy the modulus equation: $|x - 3| = 7$.

4. Solve the modulus inequality: $|2x + 3| < 9$.

5. Determine the solution set for the modulus equation: $|x^2 - 9| = 0$.

6. Solve the modulus inequality: $|4 - 2x| \leq 6$.

7. Given that $|2y - 1| = 3$, find the possible values for y .

8. Solve the system of equations and inequalities:

$$\begin{aligned}|x - 2| &\leq 3 \\ 2y + 1 &> 5\end{aligned}$$

9. Find the values of a for which $|a^2 - 4| = 0$.

Problem 4 (Triangle Inequality). Solve the following problems based on the Triangle Inequality Theorem. Show all necessary steps and reasoning.

1. State and prove the Triangle Inequality Theorem.

2. Given a triangle with sides $a = 4$, $b = 7$, and $c = 10$, determine whether it is a valid triangle.

3. For a triangle with sides x and y , express the condition under which the triangle is degenerate.

4. If the sides of a triangle are $m + 2$, $3m - 1$, and 4 , find the possible range of values for m .

5. Determine the possible lengths of the third side of a triangle with sides 8 and 15 .

6. Given a triangle with sides p and q , find the range of values for the third side such that the triangle is non-degenerate.
7. Show that in any triangle, the sum of the lengths of any two sides is greater than the length of the third side.
8. If a , b , and c are the sides of a triangle, prove that $a + b > c$.
9. A triangle has sides $a = 5$ and $b = 12$. Determine the possible range of values for the length of side c .
10. Consider a triangle with sides s and t . Express the Triangle Inequality Theorem in terms of s and t .
11. If $x + y = 10$ and $x - y = 4$, determine whether it is possible to form a triangle with sides x , y , and z .
12. Given a triangle with sides l , m , and n , prove that the longest side is opposite the largest angle.
13. Determine the condition under which the Triangle Inequality Theorem becomes an equality.
14. Find the perimeter of a triangle with sides $a = 6$, $b = 8$, and $c = 12$.
15. Consider a triangle with sides $p = 3$ and $q = 4$. If the third side is an integer, find the possible values for the third side.
16. $|a| \leq 1$ implies $|3 + a^3| \leq 4$.
17. $|b| < 1$ implies $|3 - b| \geq 2$.

Problem 5 (Number systems and conversion). 1. Are the following even or odd. (State reasons why using the correct methods.)

$$2, 14, -100, 23, -17, 4m + 6n + 1 \text{ for } m, n \in \mathbb{Z}$$

2. Convert the decimal number 243 to binary.
3. Convert the binary number 110110 to decimal.
4. Convert the octal number 753 to binary.
5. Convert the hexadecimal number 1A3 to decimal.
6. Convert the decimal number 156 to octal.
7. Convert the binary number 101101 to hexadecimal.
8. Perform the addition $1101_2 + 101_2$ in binary.

9. Perform the subtraction $10101_2 - 110_2$ in binary.
10. Perform the multiplication $37_8 \times 24_8$ in octal.
11. Perform the division $1101_2 \div 11_2$ in binary.
12. Convert the decimal number 78 to hexadecimal.
13. Convert the octal number 345 to decimal.
14. Perform the addition $1A3_{16} + C7_{16}$ in hexadecimal.
15. Perform the subtraction $2F_{16} - 14_{16}$ in hexadecimal.
16. Convert the hexadecimal number 1F8 to binary.

Problem 6 (Non-positional number systems). 1. Explain the concept of non-positional number systems. Provide examples of such systems.

2. Compare and contrast positional and non-positional number systems. Highlight advantages and disadvantages of each.
3. Convert the decimal number 42 to binary using the non-positional number system.
4. Write the Roman numeral representation for the following numbers: 16, 89, 394, and 2019.
5. Develop a set of rules for the addition of Roman numerals. Provide examples to illustrate the rules.
6. Investigate the limitations of Roman numerals in arithmetic operations. Discuss why they were not well-suited for certain mathematical tasks.
7. Convert the Roman numeral XIV to decimal form.

Problem 7 (Logic, statement and truth tables). 1. Define the terms proposition and logical statement. Provide an example of each.

2. Explain the difference between a tautology and a contradiction. Provide examples.
3. Write the negation of the following statements:
 - (a) P : The sky is blue.
 - (b) Q : $2 + 2 = 5$.
 - (c) R : It is not raining.
4. Construct truth tables for the following compound propositions:
 - (a) $P \wedge Q$
 - (b) $P \vee Q$

(c) $P \rightarrow Q$

5. Determine whether the following statements are equivalent:

(a) $(P \wedge Q) \vee R$ and $P \wedge (Q \vee R)$

(b) $P \rightarrow Q$ and $\neg P \vee Q$

6. Write the contrapositive and converse of the following implication:

$$P \rightarrow Q$$

7. Given that P is true and Q is false, determine the truth values of the following compound propositions:

(a) $P \wedge \neg Q$

(b) $\neg P \vee (P \wedge Q)$

8. Show that the following argument is valid using a truth table:

$$\frac{P \rightarrow Q \quad Q \rightarrow R}{\therefore P \rightarrow R}$$

9. Prove the following logical equivalence using truth tables:

$$(P \wedge Q) \vee \neg P \equiv Q \vee \neg P$$

10. Given the statements P : "It is sunny" and Q : "I will go for a walk," express the compound statement "If it is sunny, then I will go for a walk" using logical symbols.

11. Determine whether the statement "If it is not raining, then I will go for a run" is equivalent to its contrapositive.

12. Use De Morgan's laws to simplify the following expression:

$$\neg(P \wedge Q)$$

13. Construct a truth table for the biconditional statement:

$$P \leftrightarrow Q$$

14. Determine the truth value of the statement "Either it is not raining, or I am carrying an umbrella" if it is known that the first part is true and the second part is false.

15. Given that $P \rightarrow Q$ is true and $Q \rightarrow R$ is true, show that $P \rightarrow R$ is also true using logical reasoning.

16. Write a logical statement that is equivalent to the exclusive or (XOR) operation and provide its truth table.

Problem 8 (Compound statements). 1. Define what a compound statement is and provide an example.

2. Express the compound statement "If it is sunny, then I will go for a run and eat ice cream" using logical symbols.
3. Construct the truth table for the compound statement $p \wedge (\neg q \vee r)$.
4. Determine the negation of the compound statement: "If it is a weekday, then I have to go to work and attend a meeting."
5. Simplify the compound statement $(p \wedge q) \vee (\neg p \wedge \neg q)$.
6. Write the contrapositive of the statement: "If the car is not in the garage, then it is not safe."
7. Prove the logical equivalence of $p \vee (q \wedge r)$ and $(p \vee q) \wedge (p \vee r)$ using truth tables.
8. Convert the compound statement "I will go to the party if and only if my friends are also going" into symbolic form.
9. Use De Morgan's laws to simplify the negation of the compound statement $(p \wedge \neg q) \vee \neg(r \vee s)$.
10. Show that the statements $p \rightarrow q$ and $\neg q \rightarrow \neg p$ are logically equivalent.
11. Determine if the following argument is valid: "If it is raining, then the streets are wet. The streets are wet. Therefore, it is raining."
12. Express the compound statement "Either I will pass the exam, or I will retake the course" using logical symbols.
13. Determine the converse of the statement: "If the light is on, then the room is bright."
14. Use logical equivalences to prove that $p \vee (q \wedge r)$ is equivalent to $(p \vee q) \wedge (p \vee r)$.
15. Write the inverse of the statement: "If I study hard, then I will get a good grade."

Problem 9 (Logical equivalence). Answer the following questions regarding logical equivalence. Clearly show all steps of your reasoning.

1. Determine if the following statements are logically equivalent:
 $p \wedge (q \vee r)$ and $(p \wedge q) \vee (p \wedge r)$
2. Prove the logical equivalence of $\neg(p \wedge q)$ and $\neg p \vee \neg q$.
3. Show that $p \rightarrow (q \wedge r)$ is logically equivalent to $(p \rightarrow q) \wedge (p \rightarrow r)$.

4. Simplify the expression $(p \wedge q) \vee (\neg p \wedge r)$.
5. Determine if $p \vee (q \wedge r)$ is equivalent to $(p \vee q) \wedge (p \vee r)$.
6. Prove the logical equivalence of $p \wedge (q \rightarrow r)$ and $(p \wedge q) \rightarrow (p \wedge r)$.
7. Simplify the expression $\neg(p \vee \neg q)$.
8. Show that $(p \wedge q) \rightarrow r$ is equivalent to $\neg p \vee \neg q \vee r$.
9. Determine if $\neg(p \rightarrow q)$ is logically equivalent to $p \wedge \neg q$.
10. Prove that $p \wedge (q \vee r)$ is equivalent to $(p \wedge q) \vee (p \wedge r)$.
11. Simplify the expression $(p \rightarrow q) \wedge (\neg p \vee r)$.
12. Determine if $(p \wedge q) \rightarrow r$ is equivalent to $p \rightarrow (q \rightarrow r)$.
13. Show that $p \vee (q \wedge r)$ is equivalent to $(p \vee q) \wedge (p \vee r)$.
14. Simplify the expression $\neg(p \wedge \neg q)$.
15. Prove the logical equivalence of $p \wedge (q \rightarrow r)$ and $(p \wedge q) \rightarrow (p \wedge r)$.
16. Determine if $\neg(p \rightarrow q)$ is logically equivalent to $p \wedge \neg q$.

[Predicates and Quantifiers]

Problem 10. 1. Define a predicate $P(x)$ to represent the statement " x is a prime number." Using this predicate, express the following statements:

- (a) There exists a prime number.
 - (b) All natural numbers are prime.
2. Let $Q(x, y)$ be the predicate " x is divisible by y ." Express the statement "There exists a number that is divisible by every other number" using quantifiers.
 3. Define a predicate $R(x, y)$ to represent the statement " x is greater than y ." Express the statement "There exists a real number that is greater than every other real number" using quantifiers.
 4. Consider the predicate $S(x, y)$: " x is the parent of y ." Express the statement "Everyone has a parent" using quantifiers.
 5. Define a predicate $T(x)$ to represent the statement " x is a positive integer." Express the statement "There exists a positive integer that is not divisible by 2 or 3" using quantifiers.
 6. Let $U(x, y)$ be the predicate " x is a factor of y ." Express the statement "For every integer, there exists another integer that is a multiple of it" using quantifiers.

7. Define a predicate $V(x)$ to represent the statement " x is a solution to the equation $2x - 5 = 0$." Express the statement "There exists a real number that satisfies the equation $2x - 5 = 0$ " using quantifiers.
8. Consider the predicate $W(x, y)$: " x is parallel to y ." Express the statement "There exist two lines that are parallel" using quantifiers.
9. Define a predicate $X(x)$ to represent the statement " x is a positive rational number." Express the statement "For every positive rational number, there exists another positive rational number that is greater than it" using quantifiers.
10. Let $Y(x, y)$ be the predicate " x is a friend of y ." Express the statement "Everyone has a friend" using quantifiers.
11. Define a predicate $Z(x)$ to represent the statement " x is an odd integer." Express the statement "There exists an odd integer that is not a prime number" using quantifiers.
12. Consider the predicate $A(x, y)$: " x is perpendicular to y ." Express the statement "There exist two lines that are perpendicular" using quantifiers.
13. Define a predicate $B(x)$ to represent the statement " x is a negative real number." Express the statement "There exists a negative real number that is not the square of any real number" using quantifiers.
14. Let $C(x, y)$ be the predicate " x is a cousin of y ." Express the statement "For every person, there exists another person who is their cousin" using quantifiers.
15. Define a predicate $D(x)$ to represent the statement " x is an irrational number." Express the statement "For every irrational number, there exists another irrational number that is greater than it" using quantifiers.